Yang, Chien-Yi

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Objective

4th-year CS Ph.D. candidate at UCSD (graduating 2026), seeking full-time AI compute architect roles. Focused on hardware-software co-design for AI/ML accelerators: parameterized architecture modeling, analytical PPA engines, and mapping/scheduling optimization for tensor, transformer, and LLM workloads.

Education

University of California San Diego **September 2022 – Present**
Ph.D. in Computer Science and Engineering

National Taiwan University **September 2017 – June 2021**
Bachelor of Electrical Engineering

Research Experience

University of California San Diego **September 2022 – 2026 (anticipated)**
PhD student

- **PNMAX: Mapping-Architecture Co-Exploration for AI Accelerators:** End-to-end DSE framework for processing-near-memory AI accelerators. Built a parameterized graph-based architecture representation, a hardware-aware mapping formulation, and a bottom-up analytical **PPA** engine over processing units, buffers, DRAM banks, and interconnects. Achieved up to 32.1x speedup over cycle-accurate simulators with $R^2 > 0.97$, and discovered Pareto-optimal architecture-mapping design points for **transformer/LLM workloads**.
- **OptiPIM: MILP-Based DNN Inference Acceleration:** Developed a mixed-integer linear programming engine, maximizing DNN inference throughput on digital processing in memory (PIM); reduced execution cycles by **20x** on Samsung HBM-PIM. **ISCA'25 Best Artifact Award.**
- **Multi-Objective Co-Optimization for Analog PIM:** Noise-aware Bayesian optimization (**BoTorch**) jointly tuning software and hardware parameters under device non-idealities. Achieved **3.3x** higher constraint satisfaction, **4.28%** accuracy gain, **35.4%** power reduction, and **49x** speedup.
- **NN-Steiner: ML for EDA :** Mixed neural-algorithmic solver for the rectilinear Steiner minimum tree problem; built a **PyTorch Distributed** training system at >**98% GPU utilization**, with up to **2.99%** improvement over state-of-the-art.

Industry Experience

MediaTek Research **Dec. 2021 – June 2022**
R&D Engineer Intern *Taipei, Taiwan*

- **Distributed LLM Training Infrastructure:** Deployed a large-scale distributed training pipeline (data + model parallelism) for a T5-based LLM on a cluster of **100+ GPUs** via **PyTorch Distributed**. Achieved **30x** end-to-end throughput speedup through **PyTorch Profiler**-based critical-path optimization across compute, collective communication, and data loading.
- **RL-Based Floorplanning:** Designed an RL methodology for chip floorplanning, accelerating the design flow >**3x**.

Selected Publications

- **C.-Y. Yang**, J. Liu, Y. Pan, T. Rosing, M. Zhou, L. Josipovic. PNMAX: Mapping-Architecture Co-Exploration Framework for Processing near Memory. **MICRO'26** (Under Review).
- **C.-Y. Yang**, S. Chen, M. Zhou, F. Ponzina, R. Ayoub, P. Mercati, T. Rosing. Constraint-Scheduled Bayesian Optimization for Software-Hardware Co-Optimization on HDnn-PIM. **ESL'26**.
- J. Liu, M. Zhou, Y. Pan, **C.-Y. Yang**, L. Josipovic, T. Rosing. OptiPIM: Optimizing Processing In-Memory Acceleration Using Integer Linear Programming. **ISCA'25. [ISCA Best Artifact Award]**
- **C.-Y. Yang**, M. Zhou, F. Ponzina, S. S. Prakash, R. Ayoub, P. Mercati, M. Subedar, T. Rosing. Multi-objective software-hardware co-optimization for HD-PIM via noise-aware Bayesian optimization. **ICCAD'24**.
- Kahng, A. B., R. R. Nerem, Y. Wang, and **C.-Y. Yang**. NN-Steiner: A mixed neural-algorithmic approach for the rectilinear Steiner minimum tree problem. **AAAI'24**. (**Alphabetical Order**)
- Chang, F.C., Tseng, Y.W., Yu, Y.W., Lee, S.R., Cioba, A., Tseng, I.L., Shiu, D.S., Hsu, J.W., Wang, C.Y., **Yang, C.Y.** and Wang, R.C. Flexible chip placement via reinforcement learning: late breaking results. **DAC'22**.

Skills

Languages & Systems: C/C++, Python, Rust, Tcl; PyTorch, PyTorch Distributed
Architecture & Performance Modeling: Analytical PPA modeling, Timeloop, CiMLoop, CACTI, Accelergy-style cost models; cycle-accurate simulation methodologies
EDA: Design Compiler, Innovus, RTL synthesis & physical design